REGULAR POLYTOPES REALISED OVER ARBITRARY RINGS, AFTER GROTHENDIECK'S WORKING NOTES

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Abstract. In the *Esquisse d'un programme* Grothendieck announced that the regular polytopes make sense over any field, and even over any base ring, through universal formulae in a few parameters. Four folders of his working notes, kept at the University of Montpellier and dating from the late 1970s, contain the computations behind that announcement. We state and prove the main results they contain, completed or corrected where the pages are incomplete. Over any commutative ring, a regular polygon closes after n steps exactly when a Chebyshev-type polynomial F_n vanishes at its invariant, and in characteristic p there are regular $2p$-gons whose vertices lie on two parallel lines. In any dimension, the rotation $\sigma_k\sigma_{k+1}$ of the universal realisation by reflections satisfies $(t-1)(t^2-\alpha_k t+1)=0$, and its order is 3, 4 or 5 exactly when $\alpha_k=-1$, 0 or $\alpha_k^2+\alpha_k=1$, over every nonzero ring. Realisations by pseudo-reflections with a pinned start vector are classified over any ring by $2n+1$ parameters, with automorphism group $\mathbb{G}_m\times\mathcal{U}_\rho$, where $\mathcal{U}_\rho$ deforms $\mathbb{G}_m$ into $\mathbb{G}_a$. The icosahedron written over $\mathbb{Z}[\alpha]/(\alpha^2+\alpha-1)$ stays faithful over every nonzero ring, characteristics 2 and 5 included. Its circumscribed quadric is smooth exactly when 5 is invertible, and modulo 5 it becomes a cone whose six vertex directions fill a conic of $\mathbb{P}^2(\mathbb{F}_5)$. Finally, the combinatorial icosahedron carries five distinguished triples of edge directions on which its group acts through Alt_5, and the hemi-icosahedron is rebuilt from an oriented set of five elements. All statements are formalised in Lean.

1. Introduction

In the *Esquisse d'un programme* [3, §4], written in 1984, Grothendieck describes a project from the late 1970s: to realise the regular polyhedra, and more generally the regular maps and polytopes [1, 7], not only over the real numbers but over any field and any base ring, without lengths or angles, by universal formulae in a few parameters, the doubles of the cosines of the fundamental angles. He adds that the icosahedron specialised to a finite field remains an icosahedron, and that the characteristics 2 and 5 are the most singular. Ji [4] has recently reconstructed part of this programme from the *Esquisse* alone, and observes that his own formulae for the icosahedron are undefined in characteristics 2 and 5.

The University of Montpellier holds the mathematical papers that Grothendieck left to Jean Malgoire and publishes their facsimiles. Four of its folders contain the computations behind the passage of the *Esquisse*:

- folder 70, *Réalisations géométriques de structures combinatoires (vieilles rédactions)*: the universal formulae and the icosahedron over $\mathbb{Z}[\alpha]$;
- folder 72, *Polyèdres réguliers et hyperpolyèdres réguliers (calculs en dimension quelconque)*, dated from 1978 by the inventory: realisations by pseudo-reflections over a base;
- folder 78, *Polygones réguliers et polynômes cyclotomiques*, dated from 1977: regular polygons over a ring;
- folder 86, *Polyèdres réguliers*: the icosahedron as a combinatorial object.

This paper states and proves the main results of these folders. The statements are attributed to Grothendieck when the page states them, with the page; otherwise they are said to be *after*

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the page, in a form the edition completed or corrected, and each such change is described. The proofs are ours, except where we follow the page and say so. Every result is formalised in Lean 4 with Mathlib (Section 7).

Summary. Section 2 treats polygons: the closure criterion $u_\alpha^n = 1 \iff F_n(\alpha) = 0$ over any ring (Theorem 2.1), and the regular $2p$-gons on two parallel lines in characteristic p (Proposition 2.2). Section 3 treats the universal formulae in any dimension and the order criterion (Theorems 3.1 and 3.2). Section 4 classifies pinned realisations by pseudo-reflections over a ring and computes their automorphisms (Theorems 4.1, 4.2, 4.3), and lists the invariant subspaces over a field (Theorem 4.4). Section 5 proves that the icosahedron over $\mathbb{Z}[\alpha]$ is faithful over every nonzero ring and studies its quadric and its reduction modulo 5 (Theorems 5.1, 5.2, 5.3). Section 6 treats the combinatorial icosahedron and the hemi-icosahedron (Theorems 6.1 and 6.2).

2. Regular polygons over a ring

Folder 78 looks for a definition of the regular polygon in an affine plane over an arbitrary field or ring, without lengths or angles: an n-gon each of whose combinatorial automorphisms extends uniquely to an affinity. In the frame of three consecutive vertices (s_0, s_1, s_{-1}), pages 81 to 85 reduce such a polygon, when it is *proper*, to one parameter α, its numerical invariant, and the rotation becomes

$$u_\alpha(x, y) = (1 - y + \alpha x,\ x),$$

a bijection of k^2 with inverse $(x, y) \mapsto (y, 1 - x + \alpha y)$. Let k be a commutative ring, $\alpha \in k$, and $s_n = u_\alpha^n(0, 0)$ for $n \in \mathbb{Z}$.

Let $S_m \in \mathbb{Z}[U]$ be the renormalised Chebyshev polynomials of the second kind: $S_0 = 1$, $S_1 = U$, $S_{m+1} = US_m - S_{m-1}$ for all $m \in \mathbb{Z}$, so that $S_{-1} = 0$. Following page 109, put

$$F_{2m} = S_{m-1}, \qquad F_{2m+1} = S_m + S_{m-1} \qquad (m \in \mathbb{Z}),$$

so $F_0 = 0$, $F_1 = F_2 = 1$, $F_3 = U + 1$, $F_4 = U$, $F_5 = U^2 + U - 1$, $F_6 = U^2 - 1$, and $F_{-n} = -F_n$. For t invertible with inverse t' and $m \geq 0$, one has $F_{2m+2}(t + t')(t^2 - 1)t^m = t^{2m+2} - 1$ and $F_{2m+1}(t + t')(t - 1)t^m = t^{2m+1} - 1$, by induction from the recurrence $F_{n+4} = UF_{n+2} - F_n$.

Theorem 2.1 (Grothendieck, folder 78, pp. 109–113)**.**

(1) For every $n \in \mathbb{Z}$, in $\mathbb{Z}[U]$,

$$F_nF_{n+1} - UF_{n-1}F_n + F_{n-2}F_{n-1} = 1;$$

equivalently $A_nF_{n-1} + F_nF_{n+1} = 1$ with $A_n = F_{n-2} - UF_n$. In particular F_{n-1} and F_{n+1} are coprime.

(2) For every $n \in \mathbb{Z}$ and $(x, y) \in k^2$, with $X_n = (F_nF_{n+1})(\alpha)$,

$$u_\alpha^n(x, y) = (1 - x - y)\, s_n + x\, s_{n+1} + y\, s_{n-1}, \qquad s_n = (X_n, X_{n-1}).$$

(3) For every $n \in \mathbb{Z}$ the following are equivalent: (a) $u_\alpha^n = \mathrm{id}$; (b) $s_n = s_0$; (c) u_α^n fixes some vertex s_i; (d) $F_n(\alpha) = 0$.

Proof. (1) For $n = 2m$, the four polynomials are S_{m-1}, $S_m + S_{m-1}$, $S_{m-1} + S_{m-2}$, S_{m-2}; replacing S_m by $US_{m-1} - S_{m-2}$, the left-hand side becomes Q_{m-1}, where $Q_m = S_m^2 - US_mS_{m-1} + S_{m-1}^2$. The odd case gives Q_m in the same way. Now $Q_{m+1} = S_{m+1}(S_{m+1} - US_m) + S_m^2 = S_m^2 - S_{m+1}S_{m-1} = Q_m$, and $Q_0 = 1$: this is the Cassini identity of the S_m.

(2) The identity at rank $n + 1$, evaluated at α, gives $X_{n+1} = \alpha X_n - X_{n-1} + 1$. Call $V_n(x, y)$ the right-hand side. Then $V_0 = \mathrm{id}$, since $X_0 = X_{-1} = 0$ and $X_1 = X_{-2} = 1$, and $u_\alpha \circ V_n = V_{n+1}$, because the weights $1 - x - y$, x, y sum to 1 and each $1 - X_{j-1} + \alpha X_j$ equals X_{j+1}. Induct upwards, and downwards since u_α is bijective.

(3) (a)⇒(b) is clear. (d)⇒(a): if $F_n(\alpha) = 0$, then $X_n = X_{n-1} = 0$, hence $X_{n+1} = X_{n-2} = 1$, and (2) gives $u_\alpha^n = \mathrm{id}$. (b)⇒(d): if $(F_nF_{n+1})(\alpha) = (F_{n-1}F_n)(\alpha) = 0$, multiply the identity of (1) at α by $F_n(\alpha)$:

$$F_n(\alpha) = F_n(\alpha)\,(F_nF_{n+1})(\alpha) + \big(F_{n-2}(\alpha) - \alpha F_n(\alpha)\big)(F_{n-1}F_n)(\alpha) = 0.$$

(b)⇔(c): $u_\alpha^n(s_i) = u_\alpha^i(s_n)$ and u_α^i is injective. □

The criterion holds over every commutative ring, zero divisors included: the converse goes through the Bézout relation of (1), not through a factorisation. Over a field, Bachmann and Schmidt gave the corresponding criterion for linear circulant systems, in the form recalled by Korchmáros and Sznyi [5, (3.4)]: $x^2 - cx + 1$ divides $x^n - 1$. In terms of the Lucas sequence $U_m = U_m(\alpha, 1)$ [6], $F_{2m}(\alpha) = U_m$ and $F_{2m+1}(\alpha) = U_{m+1} + U_m$. On page 113 the first Bézout coefficient is written $F_{n-2} - UF_{n-1}$, which fails already for $n = 4$; the correct coefficient is A_n above. The page also concludes "$F_n(\alpha) = 1$" on page 112, where $F_n(\alpha) = 0$ is meant. An affinity of k^2 is determined by its values at $s_0 = (0,0)$, $s_1 = (1,0)$, $s_{-1} = (0,1)$, so a combinatorial automorphism of the polygon extends in at most one way; together with $\sigma_1 u_\alpha \sigma_1 = u_\alpha^{-1}$, where $\sigma_1(x,y) = (y,x)$, the dihedral group acts on the vertices. The quadratic function $f_\alpha(x,y) = x^2 - x + y^2 - y - \alpha xy$ of page 96 is u_α-invariant and vanishes at every vertex.

Polygons on two parallel lines. A *characteristic* of a nonzero ring k is an integer $p \geq 0$ such that $m \in \mathbb{Z}$ maps to 0 in k exactly when $p \mid m$. Pages 84–85 and 94–99 describe two polygons in characteristic p whose vertices lie alternately on two parallel lines: the one with invariant $\alpha = -2$, and the *improper* one, with rotation $v(x,y) = (1-y, -x)$. Put $t_n = v^n(0,0)$.

Proposition 2.2 (after folder 78, pp. 84, 85, 94–99)**.** *Let k be a nonzero ring of characteristic p.*

(1) $s_{2m} = (-m, m)$ and $s_{2m+1} = (1+m, -m)$ for $\alpha = -2$: the even vertices lie on $x + y = 0$, the odd ones on $x + y = 1$, and $f_{-2} = (x+y)(x+y-1)$.
(2) $u_{-2}^n = \mathrm{id}$ iff $2p \mid n$, iff $F_n(-2) = 0$; so u_{-2} has order $2p$, and $s_n = s_m$ iff $2p \mid n - m$.
(3) Three consecutive vertices are never collinear: $\det(s_{n+1} - s_n, s_{n-1} - s_n) = 1$.
(4) For the improper polygon, v^{2m} is the translation by $m(1,-1)$, $t_{2m} = (m, -m)$, $t_{2m+1} = (m+1, -m)$; v has order $2p$, three consecutive vertices are not collinear, $\sigma_1 v \sigma_1 = v^{-1}$, and the linear part of v has determinant -1, while that of u_{-2} has determinant 1.

Proof. (1) By induction both ways: $u_{-2}(-m, m) = (1+m, -m)$ and $u_{-2}(1+m, -m) = (-(m+1), m+1)$. (2) $s_{2m} = (0,0)$ iff $m = 0$ in k, iff $p \mid m$; and $s_{2m+1} = (0,0)$ would force $1 = 0$. Theorem 2.1 gives the rest. (3) For $n = 2m$ the two vectors are $(1+2m, -2m)$ and $(2m, 1-2m)$, of determinant 1; the odd case is similar. (4) By induction, $v^{2m}(x,y) = (x+m, y-m)$ and $v^{2m+1}(x,y) = (1-y+m, -x-m)$, and one argues as in (2). □

No primality or parity of p is needed. For $p = 2$, $\alpha = -2 = 0$ and one recovers the square; for $p = 0$ both rotations have infinite order and the vertices form the two staircase ∞-gons of page 85. The two polygons have the same vertex set, $s_{2m+1} = t_{2m+1}$ and $s_{2m} = t_{-2m}$: the first zigzags along the two lines in opposite directions, the second climbs them in the same direction (Figure 1). Over a field, every affinely regular polygon lies on an irreducible conic [5, Th. 2.1–2.3], and the case $c = -2$ is treated there for n prime to the characteristic, where the even and odd vertices coincide; the two-line $2p$-gons do not appear there. Here the conic $f_{-2} = 0$ is the pair of lines.

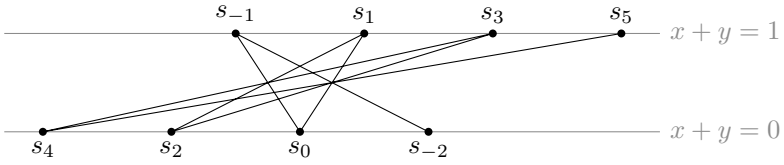


FIGURE 1. The polygon of invariant $\alpha = -2$, drawn over $\mathbb{Q}$ in the coordinates $(x - y, x + y)$. Over $\mathbb{Q}$ it does not close; in characteristic p, $s_{n+2p} = s_n$ and it is a $2p$-gon.

3. The universal formulae in any dimension

Pages 9 to 25 of folder 70 write every regular affine realisation by reflections of the polytope $\{p_1, \dots, p_n\}$ in the frame of its vertices $(s_0, \dots, s_{n+1})$, with n parameters $\alpha_0, \dots, \alpha_{n-1}$. Let R be a commutative ring and $\alpha_0, \dots, \alpha_{n-1} \in R$. In barycentric coordinates $(t_0, \dots, t_{n+1})$ with respect to this frame, put

$$\ell_i = t_i - t_{i+1} + (1+\alpha_i)(t_{i+2} + \cdots + t_{n+1}), \qquad \sigma_i = \mathrm{id} + \ell_i \otimes (s_{i+1} - s_i) \quad (0 \le i \le n),$$

and $w_i = s_{i+1} - s_i$ (for $i = n$ the coefficient α_n does not occur).

Theorem 3.1 (after folder 70, pp. 19, 23–25 and 67)**.** *Let $n \ge 1$.*

(1) For all α_i: $\sigma_i^2 = 1$; $\sigma_i\sigma_j = \sigma_j\sigma_i$ for $i+2 \le j \le n$; σ_i preserves $t_0 + \cdots + t_{n+1}$; $\sigma_i s_i = s_{i+1}$, $\sigma_i s_j = s_j$ for $j < i$, and $\sigma_i s_j = s_j + (1+\alpha_i)w_i$ for $i+2 \le j \le n+1$.

(2) For $k \le n-1$ the image of $\sigma_k\sigma_{k+1} - 1$ lies in the plane $W = \langle w_k, w_{k+1}\rangle$, and $(t-1)(t^2 - \alpha_k t + 1)$ annihilates $\rho = \sigma_k\sigma_{k+1}$.

(3) If $R \ne 0$, ρ has order exactly 3, 4 or 5 iff, respectively, $\alpha_k = -1$, $\alpha_k = 0$, or $\alpha_k^2 + \alpha_k - 1 = 0$.

Proof. (1) $\ell_i(w_i) = -2$ gives $\ell_i(\sigma_i x) = -\ell_i(x)$, hence $\sigma_i^2 = 1$. For $i+2 \le j \le n$, $\ell_i(w_j) = 0 = \ell_j(w_i)$, so both products are $x \mapsto x + \ell_i(x)w_i + \ell_j(x)w_j$. The rest is read on the values $\ell_i(s_j)$.

(2) Since $\ell_{k+1}(w_k) = 1$ and $\ell_k(w_{k+1}) = 2 + \alpha_k$,

$$\rho w_k = (1+\alpha_k)w_k + w_{k+1}, \qquad \rho w_{k+1} = -(2+\alpha_k)w_k - w_{k+1},$$

so ρ has trace α_k and determinant 1 on W, which contains the image of $\rho - 1$.

(3) Write $\rho^m x = x + \sum_{j<m} \rho^j(\rho x - x)$. On W, $\sum_{j<3} \rho^j = (1+\alpha_k)\rho$, $\sum_{j<4} \rho^j = \alpha_k((1+\alpha_k)\rho - 1)$ and $\sum_{j<5} \rho^j = (\alpha_k^2 + \alpha_k - 1)(\alpha_k\rho - 1)$, which vanish under the corresponding hypothesis. Conversely, on $\langle s_k, s_{k+1}, s_{k+2}\rangle$, $\rho(a,b,c) = \big(c - (1+\alpha)b,\ a + (1+\alpha)b,\ b\big)$ with $\alpha = \alpha_k$, and the iterates of $s_k = (1,0,0)$ are $(0,1,0)$, $(-1-\alpha, 1+\alpha, 1)$, $(-2\alpha-\alpha^2, \alpha+\alpha^2, 1+\alpha)$, then vectors whose last two coordinates are $(\alpha^3 + \alpha^2 - \alpha, \alpha + \alpha^2)$ and $\big((\alpha^2-1)P, \alpha P\big)$, with $P = \alpha^2 + \alpha - 1$. So $\rho^3 s_k = s_k$ forces $1 + \alpha = 0$; $\rho^4 s_k = s_k$ forces $\alpha + \alpha^2 = 0 = \alpha^3 + \alpha^2 - \alpha$, hence $\alpha = \alpha(\alpha + \alpha^2) - (\alpha^3 + \alpha^2 - \alpha) = 0$; and $\rho^5 s_k = s_k$ forces $P = \alpha \cdot \alpha P - (\alpha^2 - 1)P = 0$. Under each hypothesis, the iterates of order $m < p$ have a coordinate equal to 1 where s_k has 0, so $\rho^m \ne 1$ when $R \ne 0$. □

On the affine plane $s_k + W$, which ρ preserves, the linear part of ρ has trace α_k and determinant 1; over $\mathbb{R}$ it is conjugate to the rotation by $2\pi/p$ when $\alpha_k = 2\cos(2\pi/p)$, and -1, 0, $(\sqrt{5}-1)/2$ give $p = 3, 4, 5$ (Figure 2). The parameters are thus twice the cosines of the rotation angles, which is presumably the sense of the *Esquisse*'s "doubles of the cosines of the fundamental angles". The five Platonic solids are $(\alpha_0, \alpha_1) = (-1,-1)$ (tetrahedron), $(0,-1)$ (cube), $(-1,0)$ (octahedron), $(\alpha,-1)$ (dodecahedron) and $(-1,\alpha)$ (icosahedron), with $\alpha^2 + \alpha = 1$; the other root of $\alpha^2 + \alpha - 1$ gives the star polyhedra of page 69.

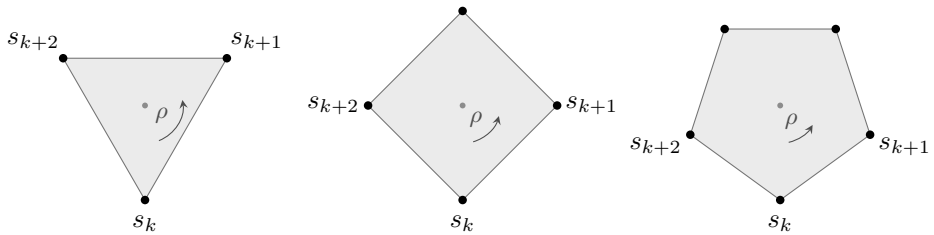


Figure 2. The orbit of s_k under $\rho = \sigma_k\sigma_{k+1}$ in the affine plane $s_k + W$, over $\mathbb{R}$, in an affine coordinate where the linear part of ρ is a rotation.

Page 67 states for $n = 2$, without proof, that $\rho^p = \mathrm{id}$ iff $F_p(\alpha) = 0$. For arbitrary p we have the following form.

Theorem 3.2 (after folder 70, p. 67)**.** *Let $n \geq 1$, $k \leq n-1$ and $\rho = \sigma_k\sigma_{k+1}$.*

(1) If $R \neq 0$, then for every $m \geq 0$, $\rho^m = 1$ iff $\chi = (t-1)(t^2 - \alpha_k t + 1)$ divides $t^m - 1$ in $R[t]$.

(2) For $p \geq 3$ let $\Psi_p \in \mathbb{Z}[x]$ be the minimal polynomial of $2\cos(2\pi/p)$, of degree d. Then $x^d\Psi_p(x + x^{-1}) = \Phi_p(x)$, the p-th cyclotomic polynomial, and $2d = \varphi(p)$.

(3) If $R = K$ is a field of characteristic 0 and $p \geq 3$, then ρ has order exactly p iff $\Psi_p(\alpha_k) = 0$.

Proof. (1) $\chi(\rho) = 0$ by Theorem 3.1(2). Conversely, if $\rho^m = 1$, the remainder $r = r_0 + r_1 t + r_2 t^2$ of $t^m - 1$ modulo the monic χ satisfies $r(\rho)s_k = r_0 s_k + r_1 s_{k+1} + r_2(s_{k+2} + (1+\alpha_k)w_k) = 0$, whence $r_2 = r_1 = r_0 = 0$: the vector s_k is cyclic. (2) $T = x^d\Psi_p(x + x^{-1})$ is an integer polynomial of degree at most $2d$ and constant term 1, vanishing at $\zeta = e^{2i\pi/p}$, so $\Phi_p \mid T$; and $\deg \Phi_p = [\mathbb{Q}(\zeta) : \mathbb{Q}(\cos(2\pi/p))]\, d \geq 2d$, since $\mathbb{Q}(\cos(2\pi/p))$ is real and ζ is not. Hence $T = \Phi_p$. (3) Let ζ be a root of $t^2 - \alpha_k t + 1$ in an algebraic closure; $\alpha_k = \zeta + \zeta^{-1}$. In characteristic 0, $t^m - 1$ has simple roots, so by (1) $\rho^m = 1$ iff $\zeta^m = 1$ and $\zeta^2 \neq 1$; thus ρ has order p iff ζ is a primitive p-th root of unity, iff $\Phi_p(\zeta) = 0$, iff $\Psi_p(\alpha_k) = 0$ by (2). □

For $p = 3, 4, 5$, $\Psi_p = x + 1$, x, $x^2 + x - 1$, and Theorem 3.1(3) is the same statement over every nonzero ring. In positive characteristic the criterion for arbitrary p is not proved here.

4. Pinned realisations by pseudo-reflections over a base

Folder 72 studies realisations of the universal regular polytope of rank $n+1$ by pseudo-reflections

$$\tau_i = \mathrm{id} + a_i' \otimes a_i \qquad (0 \leq i \leq n)$$

of a module E over a commutative ring k, with $a_i \in E$, $a_i' \in E^\vee$, and non-consecutive τ_i commuting. Given a start vector s_0, put $s_{i+1} = \tau_i s_i$ and $\nu_i = \langle s_0, a_0'\rangle\langle a_0, a_1'\rangle \cdots \langle a_{i-1}, a_i'\rangle$. Pages 96 to 100 and 103 compare two sets of conditions; a family is called *free* in the strong sense of page 103 when it is independent and spans a direct summand.

(A) non-consecutive τ_i commute, $\tau_i s_0 = s_0$ for $1 \leq i \leq n$, and $(s_0, \ldots, s_{n+1})$ is free;

(B) $\langle a_i, a_j'\rangle = 0$ for $|i-j| \geq 2$, $\langle s_0, a_i'\rangle = 0$ for $1 \leq i \leq n$, $(s_0, a_0, \ldots, a_n)$ is free, and ν_n is invertible.

Theorem 4.1 (after folder 72, pp. 96–100 and 103)**.** *Over every commutative ring k and for every k-module E, (A) and (B) are equivalent. For independence alone, (A) does not imply the invertibility of ν_n: over $\mathbb{Z}$, with $n = 0$, $s_0 = e_0$, $a_0 = e_1$, $a_0' = 2e_0^*$, the family $(s_0, s_1) = (e_0, e_0 + 2e_1)$ is independent and $\nu_0 = 2$. If $(s_0, \ldots, s_{n+1})$ is a basis of E, then E is free.*

Sketch. One reduces to the submodule N spanned by the family, in which it is a basis. Under (A), $s_{i+1} - s_i = \langle s_i, a_i'\rangle a_i$, and the coordinate on s_{i+1} of the projection onto N shows that $\langle s_i, a_i'\rangle$ is invertible; so $a_i \in N$, the τ_i restrict to N, and the commutation of τ_i and τ_j becomes $\langle a_i, a_j'\rangle = 0$. Under (B), $\langle s_i, a_i'\rangle = \nu_i/\nu_{i-1}$, and in both directions the transition matrix between $(s_0, \ldots, s_{n+1})$ and $(s_0, a_0, \ldots, a_n)$ is triangular with invertible diagonal. □

A realisation satisfying (B) with E free of rank $n+2$ is *pinned* and *normalised* when $\langle s_0, a_0'\rangle = 1$ and $\langle a_{i-1}, a_i'\rangle = 1$. Its invariants are $\mu_i = \langle a_i, a_i'\rangle = \lambda_i - 1$ and $\beta_i = \langle a_{i+1}, a_i'\rangle$, so that $\tau_i a_i = \lambda_i a_i$ and $\tau_i a_{i+1} = a_{i+1} + \beta_i a_i$.

Theorem 4.2 (after folder 72, p. 13 and pp. 91–100)**.** *A normalised pinned realisation is isomorphic, by a unique isomorphism sending s_0 to e_0 and conjugating the τ_i, to the coordinate realisation on k^{n+2} with basis $(s_0, a_0, \ldots, a_n)$ and parameters (μ_i, β_i). The coordinate realisation satisfies (B) for all parameters, and τ_i is invertible iff λ_i is. A linear map between two normalised realisations that is compatible with s_0 and the τ_i forces their parameters to be equal. Non-degenerate pinned realisations on a free module of rank $n+2$ are thus classified, rigidly, by the $2n+1$ parameters $(\lambda_0, \beta_0, \lambda_1, \ldots, \beta_{n-1}, \lambda_n)$, with $\lambda_i \in k^\times$ and $\beta_i \in k$.*

Over $\mathbb{R}$, for reflection representations of Coxeter groups, the reflections determine their Cartan matrix up to diagonal conjugation [2, §3.1], and Monson and Schulte study reflection groups of polytopes over finite fields through a similar Gram matrix [8]. Theorem 4.2 works over any ring, with pseudo-reflections of arbitrary eigenvalue λ_i, and the start vector s_0 removes the diagonal ambiguity.

Automorphisms. In the basis $(s_0, a_0, \dots, a_n)$ let π be the first coordinate, and $\tilde{c}$ the unique vector killed by all the a_i' whose coordinate on a_n is 1, obtained by back substitution. Put $\rho = \pi(\tilde{c})$; for $n = 1$, $\rho = \mu_0\mu_1 - \beta_0$, which is $4 - \beta_0$ for reflections, as on page 41. Page 37 introduces the group $\mathcal{U}_\rho = \operatorname{Spec} \mathcal{O}_S[T]_{(1+\rho T)}$ with law $\lambda * \lambda' = \lambda + \lambda' + \rho\lambda\lambda'$, acting by $u_\lambda = \mathrm{id} + \lambda\,\pi \otimes \tilde{c}$; the theorem of page 38 asserts without proof that $\mathbb{G}_m \times \mathcal{U}_\rho \to \mathrm{Aut}(E, (\tau_i))$ is an isomorphism.

Theorem 4.3 (after folder 72, pp. 37–38, on points)**.** *The $\lambda \in k$ with $1 + \rho\lambda$ invertible form a commutative group $\mathcal{U}_\rho(k)$; $\lambda \mapsto 1 + \rho\lambda$ is surjective onto the units of every k-algebra iff ρ is invertible, and $\mathcal{U}_0 = \mathbb{G}_a$. One has $u_\lambda u_{\lambda'} = u_{\lambda * \lambda'}$ and $\det u_\lambda = 1 + \rho\lambda$. An endomorphism of E commutes with all the τ_i iff it is $c + t\,\pi \otimes \tilde{c}$ for a unique (c, t), and it is invertible iff c and $c + t\rho$ are. Hence $k^\times \times \mathcal{U}_\rho(k) \cong \mathrm{Aut}(E, (\tau_i))$, $(c, \lambda) \mapsto c\,u_\lambda$. The start vectors on the line $D_0 = \bigcap_{i \geq 1} \ker a_i'$ avoiding the centre and the hyperplane $\pi = 0$, normalised by $\langle s, a_0' \rangle = 1$, are exactly the $s_0 + \lambda\tilde{c} = u_\lambda s_0$, each for a single λ.*

So $\mathcal{U}_\rho$ is a one-parameter group that is $\mathbb{G}_m$ when ρ is invertible, acting on the start line by homotheties of ratio $1 + \rho\mu$ centred at the point where π vanishes, and $\mathbb{G}_a$ when $\rho = 0$, acting by translations (Figure 3). The test for the converse is the ring of dual numbers, where $1 + \varepsilon$ is reached only if ρ is invertible.

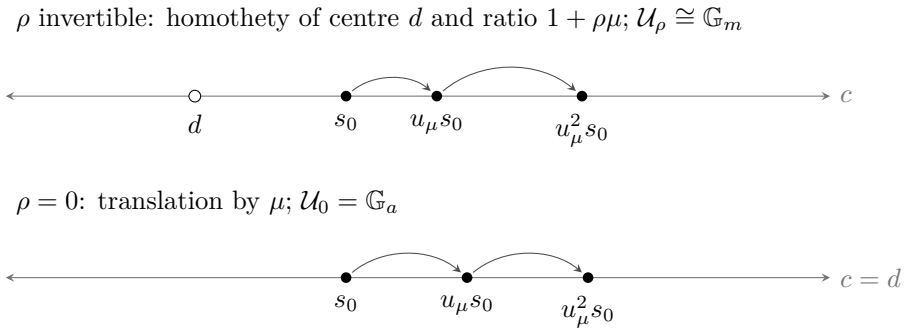


FIGURE 3. The projective line D_0 of start vectors, in the chart $\langle s, a_0' \rangle = 1$, with the centre c at infinity and d the point where π vanishes. Drawn for $\rho = 1$, $\mu = 0.6$, then $\rho = 0$, $\mu = 0.8$.

Invariant subspaces. Over a field, with $h_i = ka_i$, $H_i = \ker a_i'$, $Z_i' = H_0 \cap \cdots \cap H_i$, $Z_i^* = \langle h_{i+1}, \dots, h_n \rangle$ and C the hyperplane $\pi = 0$:

Theorem 4.4 (Grothendieck, folder 72, p. 79)**.** *Over a field, for arbitrary μ_i (characteristic 2 included), a subspace of E is stable under all the τ_i iff it is 0, E, the line $k\tilde{c}$, the hyperplane C, or Z_i' or Z_i^* for some $i < n$ with $\beta_i = 0$.*

Corollary 4.5 (after folder 72, pp. 80–82)**.** *The representation on C is irreducible iff all the β_i are nonzero and, for $n \geq 1$, $\rho \neq 0$.*

Proof. Intersect the list of Theorem 4.4 with C: $E \not\subset C$ since $\pi(s_0) = 1$; $k\tilde{c} \subset C$ iff $\rho = 0$; $Z_i^* \subset C$ always. If $\rho = 0$ and $n \geq 1$, $k\tilde{c}$ is a line of C different from C, since $h_0 \subset C$ is not a multiple of $\tilde{c}$, whose last coordinate is 1. □

The struck-out (d) of page 80 asks for $\rho \neq 0$ and all $\beta_i \neq 0$; the (c) of page 82 omits $\rho \neq 0$, which is needed. For $i < n$, $Z_i^* \subset Z_i'$ iff $\beta_i = 0$, the "$\in$" of the margin of page 79 rather than the "$\notin$" of page 81.

5. The icosahedron over $\mathbb{Z}[\alpha]$

Section VII of folder 70 writes the regular icosahedron over $\mathbb{Z}[\alpha] = \mathbb{Z}[x]/(x^2+x-1)$ by three reflections in the vertex frame $(s_0, s_1-s_0, s_2-s_0, s_3-s_0)$ (pp. 74–75). Let R be a commutative ring with $\alpha^2+\alpha-1=0$, put $\alpha' = -1-\alpha$ (the other root), and

$$\sigma_0(x,y,z) = \big(1-(x+y+z), y, z\big), \quad \sigma_1(x,y,z) = (y+\alpha' z, x-\alpha' z, z), \quad \sigma_2(x,y,z) = (x,z,y),$$

$\iota(x,y,z) = (\alpha'-x, 1-\alpha'-y, 1-\alpha'-z)$. The twelve vertices are $s_0=0$, $s_1=e_1$, $s_2=e_2$, $s_3=e_3$, $s_3'=(\alpha',-\alpha',1)$, $s_3''=(0,-\alpha',1)$ and their images by ι. Put

$$q_0 = x^2+y^2+z^2-\alpha yz+zx+xy, \qquad q = q_0-(x+y+z).$$

The thirty edges are the pairs of vertices at q_0-distance 1 over $\mathbb{Q}(\sqrt{5})$, and the twenty faces are the triangles of edges. A point p is *on the line* (r,u) if $p-r = a(u-r)$ for some $a \in R$, and *in the plane* (r,u,w) if $p-r = s(u-r)+t(w-r)$; over a field these are collinearity and coplanarity.

Theorem 5.1 (after folder 70, pp. 74–94)**.** *Let R be a nonzero commutative ring in which $\alpha^2+\alpha=1$.*

1. *The σ_i permute the twelve vertices, by permutations of $\{0,\dots,11\}$ independent of R; the vertices lie on $q=0$, and $q_0(s'-s)=1$ for each edge $\{s,s'\}$.*
2. *The twelve vertices are distinct; no three distinct vertices are collinear; and no vertex lies in the plane of a face not containing it.*
3. *The group G generated by $\sigma_0,\sigma_1,\sigma_2$ has order 120 and acts faithfully on the vertices.*
4. *If $2=0$ in R, ι is the translation by (α',α,α).*

Proof. (1) and (4) are polynomial identities modulo $\alpha^2+\alpha-1$. For (2), the σ_i are affine, so a word in them transports equality of vertices, membership of a vertex in the line of two others, and in the plane of three others, along the corresponding permutation. Pages 93–94 invoke this homogeneity and treat only base cases; we make it explicit by two tables. For every ordered pair of distinct indices, a word sends it to (s_0,s_1), (s_0,s_3'') or $(s_0,\iota s_0)$; for every face, a word sends it to (s_0,s_1,s_2). These $132+20$ facts concern permutations of twelve points and are checked by computation. In the base cases, s_k is on the line (s_0,s_j) only if the three 2×2 minors of (s_k,s_j) vanish; for $j\in\{1,5,6\}$ and each of the thirty indices $k\neq 0,j$, one of them is a unit of $\mathbb{Z}[\alpha]$ (±1, $\pm(1+\alpha)$, $\pm(2+\alpha)$ or $\pm(3+2\alpha)$), which gives $1=0$. A vertex in the plane $z=0$ of the face (s_0,s_1,s_2) would have third coordinate 0, whereas the nine other vertices have third coordinate 1, $2+\alpha$ or $1+\alpha$, all units. For (3), a breadth-first enumeration gives 120 words in the three permutations, closed under left multiplication by the generators; and an element of G is affine and determined by the images of $s_0,\dots,s_3$, so G is isomorphic to its permutation group. □

Ji [4] writes formulae over $\mathbb{Z}[\cos\theta,\cos\gamma]$ whose reduction is undefined at 2 and 5, shows that the automorphism group stays A_5 at the other primes, and notes that Grothendieck "seems to claim this in all cases". Theorem 5.1 confirms the claim, for the incidences, in every characteristic, by the integral frame of the page. The proof needs no field and no norm computation beyond thirty invertible minors. The margin "c'est en fait faux en car. 2 !" therefore concerns neither the non-collinearity nor the face planes, and the translation vector of ι in characteristic 2 is (α',α,α), not the $(\alpha',1,1)$ read on page 91.

Theorem 5.2 (Grothendieck, folder 70, pp. 101–106)**.** *Let R be a commutative ring with $\alpha^2+\alpha=1$.*

1. *q is invariant under the σ_i, and every quadratic function vanishing at the nine vertices $s_0,s_1,s_2,s_3,s_3',s_3'',\iota s_0,\iota s_1,\iota s_2$ is a multiple of q.*
2. *The half-discriminant of q_0 is 1.*
3. *The projective quadric $Q = q_0(x,y,z)-(x+y+z)t$ has discriminant $-\alpha-3$, of norm 5; it is smooth iff 5 is invertible in R.*

(4) If $5 = 0$ *and* $\alpha = 2$*, Q is a cone with vertex the centre* $o = (1, 2, 2)$*. Over a field, Q is singular iff the characteristic is* 5*.*

Proof. The uniqueness follows from a 9×9 minor equal to $13\alpha + 21$, a unit of $\mathbb{Z}[\alpha]$ with inverse $13\alpha - 8$; it holds over any ring, without dividing. The half-discriminant $4abc + uvw - au^2 - bv^2 - cw^2$ of q_0 is $2 - (\alpha^2 + \alpha) = 1$, the *merveille* of page 101, also in characteristic 2. The 4×4 polar matrix of Q has determinant $\alpha^2 - 4 = -\alpha - 3$; since $(-\alpha - 3)(\alpha - 2) = 5$ and $(-\alpha - 3)^2 = 5(\alpha + 2)$, it is invertible iff 5 is. In characteristic 5, $\alpha^2 + \alpha - 1 = (\alpha - 2)^2$, and the kernel is spanned by $(1, 2, 2, 1)$, which lies on Q. □

Theorem 5.3 (after folder 70, p. 78)**.** *Over* $\mathbb{F}_5$*, where* $\alpha = 2$*, put* $\bar{s} = s - o$ *for a vertex s.*

(1) The twelve $\bar{s}$ *are isotropic for* q_0*, and the nonzero isotropic vectors of* q_0 *are exactly the twelve* $\bar{s}$ *and their doubles* $2\bar{s}$*, twenty-four distinct vectors.*

(2) A point of $\mathbb{P}^2(\mathbb{F}_5)$ *lies on the conic* $q_0 = 0$ *iff it is one of the six vertex directions.*

(3) The six vertex directions, fifteen edge directions $s' - s$ *and ten face directions* $\bar{a} + \bar{b} + \bar{c}$ *are the* $31 = 5^2 + 5 + 1$ *points of* $\mathbb{P}^2(\mathbb{F}_5)$*, each once.*

(4) A face direction is q_0-orthogonal to the plane of its face. Through an edge direction pass exactly two tangents to the conic, through a face direction none; q_0 *takes the values* 1 *or* 4 *on* 60 *vectors and* 2 *or* 3 *on* 40*, so that* $125 = 1 + 24 + 60 + 40$*.*

Proof. All statements concern the 125 vectors of $\mathbb{F}_5^3$ and finite lists of vertices, edges and faces, and are checked by computation. □

Page 78 gives the table of directions, the count $125 = 1 + 24 + 60 + 40$, and the phrase "deux icos.": the twenty-four isotropic vectors are the vertices of the icosahedron and of its homothetic image of ratio 2. Points (2) and (4) explain the fifteen edge directions as the exterior points of the conic and the ten face directions as the interior points; that $\mathfrak{A}_5 \cong \mathrm{PSL}_2(\mathbb{F}_5)$ acts on the six axes as on $\mathbb{P}^1(\mathbb{F}_5)$ is classical, but we have not found the icosahedron drawn as a conic in $\mathbb{P}^2(\mathbb{F}_5)$ in print. This is the most singular characteristic of the *Esquisse*: the quadric degenerates to a cone, and the icosahedron survives on its isotropic directions.

6. The combinatorial icosahedron and the hemi-icosahedron

Folder 86 takes the icosahedron as a purely combinatorial object. Label its vertices $0, \dots, 11$, with 0 and 11 opposite and $1, \dots, 5$ and $6, \dots, 10$ the two pentagons of their neighbours; everything below is computed from the adjacency. For a face $\{x, y, z\}$, let $\mathrm{o}(x, y, z)$ be the third vertex of the other face bordering $\{x, y\}$, and let ιx be the antipode, the only vertex that is neither x nor a neighbour of x nor a neighbour of a neighbour. A *flag* is an oriented triangle (s, t, u); there are 120. Two edges have the same *direction* if they are equal or antipodal. For an edge $\{s, t\}$ with faces $\{s, t, u\}$, $\{s, t, v\}$, page 18 defines $\rho_0(s, t) = \{\mathrm{o}(s, u, t), \mathrm{o}(s, v, t)\}$: the side of the pentagon around s opposite to t.

Theorem 6.1 (Grothendieck, folder 86, pp. 18–25)**.**

(1) ρ_0 induces a well-defined fixed-point-free permutation ρ of the fifteen edge directions with $\rho^3 = \mathrm{id}$*. Its five orbits, the* distinguished trihedra*, each partition the twelve vertices into six edges, and at every vertex the five edges lie in the five different trihedra.*

(2) The automorphism group G of the icosahedron has order 120*, commutes with ι and ρ, and acts on the set T of the five trihedra through a homomorphism $G \to \mathfrak{S}_T$ with image* Alt_T *and kernel* $\{1, \iota\}$*.*

Proof. An automorphism commutes with o, since $\mathrm{o}(x, y, z)$ is the only common neighbour of x and y other than z, and the twelve vertices are obtained from $0, 1, 2$ by nine applications of o; so an automorphism is determined by the image of the flag $(0, 1, 2)$, and every flag is such an image. ι and ρ are defined by adjacency, hence commute with G. The rest is a computation on the 120 flags: the induced permutations of T are even, there are 60 of them, and only $(0, 1, 2)$ and $(\iota 0, \iota 1, \iota 2)$ give the identity. □

Page 23 puts two edges of different trihedra in the same triple; the correction is part of the proof's check. The page's own argument goes through the parity of the generators and the orders 6, 10 and 4 of the images of the stabilisers of a face, a vertex and an edge; here the statement is obtained by counting flags. In Euclidean terms the trihedra are the five orthogonal frames, or the five octahedra inscribed in the icosahedron; that identification is not used.

Theorem 6.2 (after folder 86, pp. 21, 26–32)**.** *To a flag (s,t,u), with $v=\mathrm{o}(s,t,u)$, associate the numbering $\nu(s,t,u)\colon\{1,\dots,5\}\to T$ formed by the trihedra of the five edges at s, read around the pentagon from t, away from u.*

(1) All the $\nu(s,t,u)$ have the same parity; they define an orientation ϖ of T independent of all choices. Every numbering compatible with ϖ is some $\nu(s,t,u)$, and $\nu(s,t,u)=\nu(s',t',u')$ iff the flags are equal or antipodal. The 60 flags of the hemi-icosahedron $\bar\Pi=\Pi/\iota$ are thus the 60 numberings of T compatible with ϖ.

(2) Changing the vertex, the edge or the face of a flag composes ν on the right with $\tau_0=(2\,4)(3\,5)$, $\tau_1=(1\,5)(2\,4)$, $\tau_2=(2\,5)(3\,4)$.

(3) Two flags of $\bar\Pi$ have the same vertex iff their numberings trace the same pentagon on T, the same edge iff they give the same partition $\{a_1\}\sqcup\{a_2,a_3\}\sqcup\{a_4,a_5\}$, and the same face iff they give the same three-element set $\{a_1,a_3,a_5\}$. One obtains 6 pentagons, all 15 partitions of type $(2,2,1)$, and the 10 three-element subsets of T.

Proof. By Theorem 6.1, $\nu(g0,g1,g2)$ is the permutation of T induced by g, composed with $\nu(0,1,2)$, which is even with our labelling; surjectivity onto Alt_T and the kernel $\{1,\iota\}$ give (1). The changes of flag are right multiplications by three fixed automorphisms, and (2) reduces to three equalities of permutations. For (3), one reduces by an automorphism to the flag $(0,1,2)$ and checks the 120 flags; the ten symmetries of the pentagon $1\,2\,3\,4\,5$ are even permutations of the places, which is why a numbering traces the pentagon of a vertex iff it is compatible with ϖ. □

The hemi-icosahedron has six vertices, fifteen edges and ten faces, its skeleton is K_6 and its group is Alt_5. The theorem rebuilds it from an oriented set of five points alone: vertices are the six pentagons compatible with the orientation, edges the fifteen partitions of type $(2,2,1)$, faces the ten three-element subsets. Pages 31–32 give the three involutions on numberings with a σ_1 that is $\tau_2\tau_1\tau_2$, conjugate to the one above; the two triples define isomorphic sets with operators.

7. Verification

Every theorem above is formalised in Lean 4 with Mathlib, in the files `Folder70.lean`, `Folder72.lean`, `Folder78.lean` and `Folder86.lean` of the repository Commutator-IO/grothendieck-project (directory `lean/Grothendieck`); the proofs use only Lean's standard axioms, and the finite statements (the words of Theorem 5.1, the computations over $\mathbb{F}_5$, the 120 flags of Section 6) are checked by the kernel. The main declarations are `F_bezout`, `u_zpow_apply`, `zpow_eq_one_iff`, `strip_vertex` and `strip_lines` in `Folder78`; `orderOf_rho_eq_three`, `orderOf_rho_eq_four`, `orderOf_rho_eq_five`, `orderOf_rho_eq_iff`, `not_collinear_any`, `not_in_face_plane_any`, `card_icoGroup`, `quadric_unique`, `qHom_smooth` and `conic_iff` in `Folder70`; `condAs_iff_condBs`, `pinned_classification`, `centraliserEquiv`, `invariant_iff` and `irreducible_on_C_iff` in `Folder72`; `rho_cube`, `permT_surjective`, `permT_eq_one_iff`, `numb_eq_iff`, `incidence_dictionary` and `counts` in `Folder86`. These files are also part of the Palomar Registry repository Commutator-IO/grothendieck-project-palomar.

Not proved here: the passage from the definition of a regular realisation to the normal form of pages 7–18 of folder 70; the order criterion for arbitrary p in positive characteristic; the identification of G with the Coxeter group $H_3\cong\mathfrak{A}_5\times\mathbb{Z}/2$; the scheme $\mathcal{U}_\rho$ and the projective

statement of folder 72 over a base; the uniqueness, suggested in the reading of folder 78, of the improper $2p$-gon among non-unimodular regular polygons.

Statement on the use of artificial intelligence

The transcriptions of folders 70, 72, 78 and 86, the modernised readings, the proofs, the Lean formalisation and a first draft of this text were produced with language models (Claude Opus 5 and Opus 5.5, Anthropic), under the author's direction. The author checked the statements, the proofs and the references, and takes full responsibility for the content. The formal proofs are checked mechanically by Lean.

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